Texas State Employee Salary Prediction Using Machine Learning

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Abstract:-The Texas State Employee Salary Prediction project was carried out during a 2-Month Summer Internship at SkillDzire and applies supervised regression to public-sector payroll data, since transparent and well-understood salary information helps government teams plan budgets and review pay structures. The objective of the project is to analyse the payroll records of Texas state employees, identify high-earning outliers and pay patterns, and build a model that predicts annual salary from employee attributes. The project uses the Texas state salary dataset obtained from the state comptroller under the Texas Public Information Act, which holds 149,481 records from 113 agencies with attributes such as agency, class title, ethnicity, gender, employment status, employ date, hourly rate, hours per week, monthly pay and annual pay. The workflow begins with dataset inspection and checks for missing values and duplicate rows, neither of which were found in the core columns, followed by exploratory data analysis. The exploratory analysis shows that the exempt regular full-time (ERF) and unclassified regular full-time (URF) statuses earn the most, with ERF averaging about $155,908 per year; that male employees average about $54,459 against $47,899 for female employees; and that Asian employees have the highest average annual pay. Preprocessing then removes the non-analytical columns, splits the employ date into year, month and day, one-hot encodes ethnicity, gender and status, and removes outliers with the IQR rule on monthly and annual pay, which drops 12,885 records and leaves 136,596. The data are split 80:20 with a fixed random state into 109,276 training and 27,320 testing records and scaled using MinMaxScaler. A Linear Regression model is trained and evaluated with MAE, MSE, RMSE and R-squared. The model reaches an R-squared of 1.00 on both the training and the testing data, with an error close to zero. This near-perfect fit arises because monthly pay was kept as an input and annual pay is exactly twelve times monthly pay, so the result reflects data leakage and does not show real predictive skill. Removing monthly pay and hourly rate from the inputs, and comparing several regression models, is therefore the necessary next step. The project gives practical exposure to the end-to-end data science workflow, from data cleaning to model evaluation, and is intended as an academic demonstration only.

TEXAS STATE EMPLOYEE SALARY PREDICTION USING MACHINE LEARNING

Sir Padampat Singhania University

SkillDzire
Data Science & Data Engineering Intern | 22-04-2026 to 22-06-2026

Objectives

- **Objective 1:** To analyse Texas state payroll records through data inspection, cleaning and exploratory data analysis.
- **Objective 2:** To build and evaluate a regression model that predicts annual salary.
- **Objective 3:** To identify high-earning outliers and pay patterns by status, gender and ethnicity.

Research Gap

- Public salary data is mostly shared as raw tables and simple summaries.
- This project joins payroll exploration with a predictive model in one notebook workflow.

Methodology

Data Preparation

- Texas salary dataset: 149,481 records, 113 agencies, 21 columns.
- No missing values in core columns; 0 duplicate rows.

EDA

- Distribution plots, scatter plots and bar plots of annual pay against agency, ethnicity, gender and status.

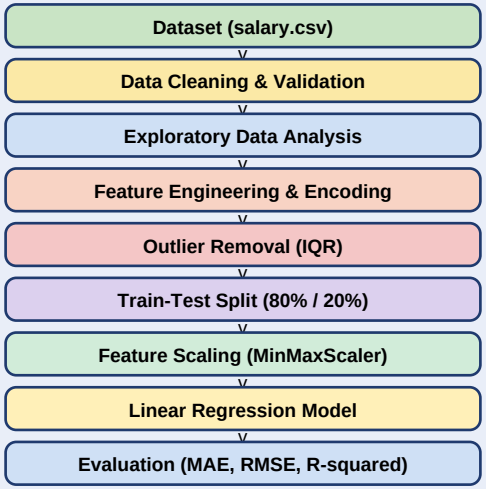
Preprocessing

- Dropped names, class codes and agency name; split employ date into year, month, day.
- One-hot encoding of ethnicity, gender, status.
- IQR outlier removal on monthly and annual pay: 136,596 records kept.
- 80:20 split (109,276 train / 27,320 test), random_state = 42; MinMaxScaler fit on training data only.

Model

- Linear Regression; MAE, MSE, RMSE, R-squared.

Flowchart / Flowdiagram



Results

Model: Linear Regression (test set, 27,320 records)

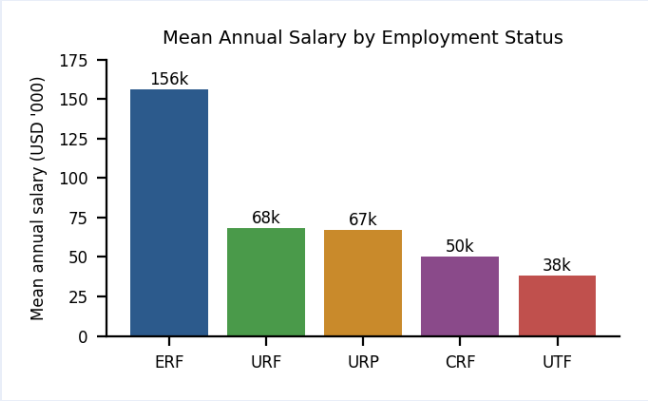
1.00 R-squared (test)	**1.00** R-squared (train)
~0.0000000002 MAE (USD)	**~0.0000000002** RMSE (USD)

Annual pay equals 12 x monthly pay and monthly pay was an input, so this score reflects data leakage, not true predictive skill.

Tools and Technology

Python | Pandas | NumPy | Matplotlib | Seaborn | SciPy | Scikit-learn | Jupyter Notebook

Key Findings from EDA



ERF and URF earn the most. Average annual pay: male $54,459, female $47,899; Asian highest by ethnicity.

Limitations

- Monthly pay was kept as a model input, so annual pay is exactly derivable (data leakage).
- Only one model (Linear Regression) was evaluated; no model comparison or cross-validation.
- Manager vs employee wage gaps and trends over time were noted but not quantified.
- Academic demonstration only.

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